

PYTHON

SQL ASSISMENT

- Write SQL query to solve the problem

City Table

```
CREATE TABLE city (  
    city_id INT PRIMARY KEY,  
    city_name VARCHAR(100) NOT NULL,  
    latitude DECIMAL(9,6),  
    longitude DECIMAL(9,6),  
    country_id INT,  
    CONSTRAINT fk_city_country  
        FOREIGN KEY (country_id)  
        REFERENCES country(country_id)  
) ENGINE=InnoDB;
```

INSERT INTO city VALUES

```
INSERT INTO city (city_id, city_name, latitude, longitude, country_id) VALUES  
(1, 'Berlin', 52.520000, 13.405000, 1),  
(2, 'Zagreb', 45.815000, 15.981900, 2),  
(3, 'New York', 40.712800, -74.006000, 3);
```

Customer Table

```
CREATE TABLE customer (  
    customer_id INT PRIMARY KEY,  
    city_id INT,  
    customer_address VARCHAR(255),  
    next_call_date DATE,  
    inserted_datetime DATETIME,
```

```

        CONSTRAINT fk_customer_city

        FOREIGN KEY (city_id)

        REFERENCES city(city_id)

    ) ENGINE=InnoDB;

```

INSERT INTO customer VALUES

```

INSERT INTO customer (customer_id, city_id, customer_address, next_call_date, inserted_datetime) VALUES

(101, 1, 'Berlin Street 1', '2025-01-10', '2024-12-01 10:30:00'),

(102, 2, 'Zagreb Center', '2025-01-15', '2024-12-02 11:00:00'),

(103, 3, 'NY Manhattan', '2025-01-20', '2024-12-03 12:00:00');

```

Country Table

```

CREATE TABLE country (

    country_id INT PRIMARY KEY,

    country_name_en VARCHAR(100) NOT NULL,

    state VARCHAR(100),

    country_code VARCHAR(10)

) ENGINE=InnoDB;

INSERT INTO country (country_id, country_name_en, state, country_code) VALUES

(1, 'Germany', 'Bavaria', 'DE'),

(2, 'Croatia', 'Zagreb County', 'HR'),

(3, 'United States', 'USA', 'US'),

(4, 'Spain', 'Madrid', 'ES'),

(5, 'Russia', 'Moscow', 'RU');

```

Task 1

List all Countries and customers related to these countries. For each country displaying its name in English, the name of the city customer is located in as well as the name of the customer. Return even countries without related cities and customers.

Task 1

```

SELECT

    co.country_name_en AS country,

    ci.city_name AS city,

    cu.customer_address AS customer_address

FROM country co

LEFT JOIN city ci

    ON co.country_id = ci.country_id

LEFT JOIN customer cu

    ON ci.city_id = cu.city_id

ORDER BY co.country_name_en;

```

Country	City	Customer Address
Croatia	Zagreb	Ivan Horwat
Germany	Berlin	Maria Smith
Russia	NULL	NULL
Spain	NULL	NULL
USA	New York	John Doe

The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search. The main editor displays a SQL query with line numbers 1 through 13. A tooltip is visible over line 2, stating: "Execute the selected portion of the script or everything, if there is no selection". The query is as follows:

```

1 • SELECT
2     Execute the selected portion of the script or everything, if there is no selection
3     ci.city_name AS City,
4     cu.customer_name AS Customer
5 FROM country co
6 LEFT JOIN city ci ON co.country_id = ci.country_id
7 LEFT JOIN customer cu ON ci.city_id = cu.city_id
8 ORDER BY co.country_name, ci.city_name;
9
10
11
12
13

```

Below the editor, the "Result Grid" tab is active, showing a table with 4 columns: Country, City, and Customer. The table contains 5 rows of data, including NULL values for Russia and Spain. The interface also includes a "Filter Rows" field, an "Export" button, and a "Wrap Cell Content" checkbox.

Country	City	Customer
Croatia	Zagreb	Ivan Horvat
Germany	Berlin	Maria Smith
Russia	NULL	NULL
Spain	NULL	NULL
USA	New York	John Doe

Task 2

Return the list of all countries that have pairs(exclude countries which are not referenced by any city). For such pairs return all customers. Return even pairs of not having a single customer

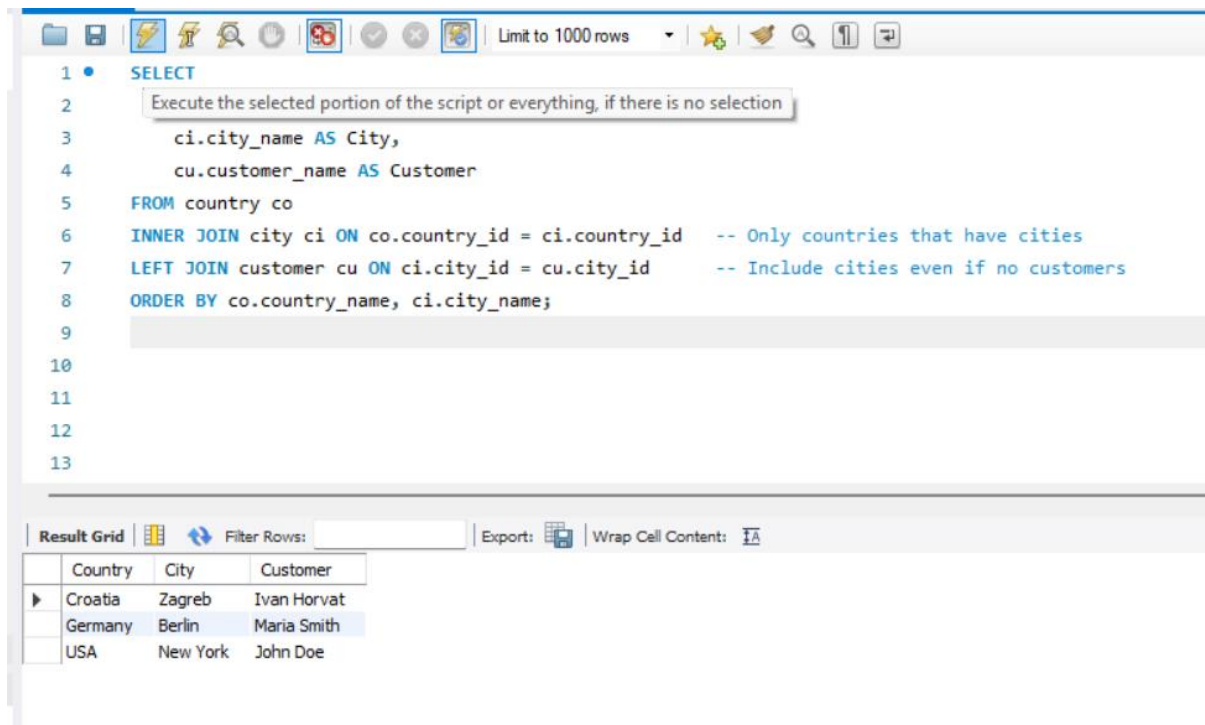
Task 2

Country City Customer Address

Croatia Zagreb Ivan Horvat

Germany Berlin Maria Smith

USA New York John Doe



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and a 'Limit to 1000 rows' dropdown. The SQL editor contains the following query:

```
1 • SELECT
2   Execute the selected portion of the script or everything, if there is no selection
3   ci.city_name AS City,
4   cu.customer_name AS Customer
5 FROM country co
6 INNER JOIN city ci ON co.country_id = ci.country_id -- Only countries that have cities
7 LEFT JOIN customer cu ON ci.city_id = cu.city_id -- Include cities even if no customers
8 ORDER BY co.country_name, ci.city_name;
9
10
11
12
13
```

Below the editor is the 'Result Grid' section. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The results are displayed in a table with three columns: Country, City, and Customer.

Country	City	Customer
Croatia	Zagreb	Ivan Horvat
Germany	Berlin	Maria Smith
USA	New York	John Doe